

ExoHabitAI

Deployment Documentation

GitHub + Render.com Cloud Deployment Guide

Live URL: <https://habitability-prediction-ml-11.onrender.com>

1. Deployment Overview

ExoHabitAI was deployed using a two-platform approach. The source code, frontend files, and compressed data files are hosted on GitHub. The live web application runs on Render.com, a cloud platform that automatically builds and deploys from the GitHub repository whenever new code is pushed. The trained XGBoost model (460KB) is hosted separately on Google Drive and downloaded at startup due to Render's filesystem constraints.

Component	Platform	Details
Source Code	GitHub	github.com/anusreechalla09-gif/HABITABILITY-PREDICTION-ML-
Web Application	Render.com	habitability-prediction-ml-11.onrender.com
ML Model (.pkl)	Google Drive	Downloaded at startup via urllib
Dataset (.csv)	GitHub (zipped)	Extracted at startup via zipfile

2. GitHub Repository Setup

2.1 Repository Structure

The repository follows a clean separation between backend (Python/Flask) and frontend (HTML/CSS/JS). Large files were compressed before uploading to stay within GitHub's file size limits.

```
HABITABILITY-PREDICTION-ML-/
  backend/
    app.py           Flask REST API
    utils.py         Preprocessing pipeline
    requirements.txt  Python dependencies (for Render)
    models/
      xgboost.zip     Compressed model (460KB)
    data/
      habitability_ranked.zip  Compressed dataset (71MB -> zipped)
      sample_planets.json     12 curated sample planets
      feature_cols.json       304 model feature names in order
  frontend/
    index.html        Main UI (3 pages)
    style.css         Space glassmorphism styles
```

```
script.js
requirements.txt
```

```
API integration & logic
Root level (ignored by Render)
```

2.2 Git Commands Used

The following commands were run from the project folder in the terminal to push the project to GitHub:

```
git init
git add .
git commit -m "Initial commit - ExoHabitAI project"
git remote add origin https://github.com/anusreechalla09-gif/HABITABILITY-PREDICTION-ML-.git
git branch -M main
git push -u origin main
```

After the initial push, subsequent updates are made using:

```
git add .
git commit -m "Update message here"
git push
```

2.3 File Size Handling

GitHub has a 100MB hard limit and warns above 50MB. Two files required special handling:

- xgboost.pkl (460KB) — GitHub blocked direct upload of .pkl files for security reasons. Solution: uploaded to Google Drive and downloaded at runtime via code
- habitability_ranked.csv (71MB unzipped) — Too large for reliable GitHub upload. Solution: compressed to habitability_ranked.zip and stored in the repository. The code extracts it at startup

3. Render.com Deployment

3.1 Creating the Web Service

To deploy on Render.com, a new Web Service was created and connected to the GitHub repository. Render automatically detects new commits and triggers a fresh build and deploy.

- Go to dashboard.render.com and click New > Web Service
- Connect GitHub account and select the HABITABILITY-PREDICTION-ML- repository
- Choose the main branch for automatic deployments
- Set the service type to Web Service with Python environment

3.2 Render Settings

The critical settings that made the deployment work correctly are:

Setting	Value	Why It Is Needed
Root Directory	backend	app.py is inside backend/ — without this Render looks at root and cannot find the

		app
Build Command	pip install -r requirements.txt	Installs all Python packages listed in backend/requirements.txt
Start Command	gunicorn app:app	Runs Flask using Gunicorn WSGI server — required for production (not development server)
Environment	Python 3	Auto-detected by Render from the project files

3.3 requirements.txt (inside backend/)

This file must be located inside the backend/ folder since Root Directory is set to backend. It lists all Python packages Render installs during the build step:

```
flask
flask-cors
xgboost
scikit-learn
pandas
numpy
joblib
gunicorn
```

Note: gunicorn is essential — without it Render cannot start the Flask application in production mode.

4. Model File Strategy

4.1 The Problem

Render's free tier has a read-only filesystem after deployment. This means files written or extracted during runtime do not persist between server restarts. The initial approach of storing xgboost.zip in GitHub and extracting it at startup failed because the extracted xgboost.pkl was lost on every restart.

4.2 The Solution — Google Drive Download

The xgboost.pkl file was uploaded to Google Drive with public sharing enabled. The utils.py code now downloads it directly from Google Drive every time the server starts:

```
GDRIVE_FILE_ID = "1CfD0hebCkQtQLOX7g6w1Z9xMJ8t-Wgok"
url = f"https://drive.google.com/uc?export=download&id={GDRIVE_FILE_ID}"
urllib.request.urlretrieve(url, MODEL_PATH)
```

After downloading, the file size is checked to confirm it downloaded correctly (expected ~459,628 bytes). If the size is too small, it means Google Drive returned an error page and the file is deleted to prevent a corrupt model from loading.

4.3 CSV Dataset Strategy

The `habitability_ranked.csv` (71MB unzipped) is stored as a zip file in GitHub. The `zipfile` module in Python's standard library extracts it at startup. Since the CSV is only used for the `/api/rank` and `/api/sample` endpoints (not for predictions), performance was optimised by loading only the first 5,000 rows for rankings and 500 rows for samples using `pandas nrows` parameter.

5. Deployment Errors & Fixes

Error Message	Root Cause	Fix Applied
ModuleNotFoundError: No module named 'app'	Root Directory was not set — Render was looking for <code>app.py</code> at the repository root instead of inside <code>backend/</code>	Set Root Directory = backend in Render Settings
SyntaxError: '[' was never closed — line 171	A list comprehension line was accidentally cut short while editing <code>utils.py</code> directly on the GitHub web editor	Replaced the entire <code>utils.py</code> file with a fresh correct version
IndentationError: unindent does not match — line 171	Mixed indentation from copy-pasting into GitHub web editor	Replaced entire <code>utils.py</code> fresh — avoided GitHub web editor for code
FileNotFoundError: <code>xgboost.pkl</code>	Render's read-only filesystem — <code>unzip</code> wrote the file but it disappeared on next restart	Switched to Google Drive download approach in <code>utils.py</code>
<code>requirements.txt</code> not found during build	The file existed only at repository root but Render looks inside Root Directory (<code>backend/</code>)	Created a new <code>requirements.txt</code> file directly inside the <code>backend/</code> folder
Prediction failed: invalid input	<code>validateForm()</code> in <code>script.js</code> was checking <code>planet_name</code> as a numeric field causing all form submissions to fail validation	Added <code>planet_name</code> to the exclusion filter in <code>validateForm()</code>

6. Live Application

Live URL:

<https://habitability-prediction-ml-11.onrender.com>

Page / Endpoint	URL Path	Status
Home Page	<code>/</code>	Live
Predictor Page	<code>/#predictor</code>	Live
Docs Page	<code>/#docs</code>	Live
Health Check	<code>/api/health</code>	Live
Single Predict	<code>/api/predict</code>	Live
Batch Predict	<code>/api/predict/batch</code>	Live
Rankings	<code>/api/rank?n=10</code>	Live

Sample Data	/api/sample	Live
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7. Full Technology Stack

Layer	Technology	Version	Purpose
Data Source	NASA Exoplanet Archive	—	33,799 exoplanet records
Data Processing	pandas + numpy	2.0 / 1.24	Feature engineering and cleaning
ML Framework	scikit-learn	1.3	Preprocessing, scaling, evaluation
Model	XGBoost	2.0	Binary habitability classifier
Model Saving	joblib	1.3	Serialize and load .pkl model
Backend	Flask	3.0	REST API server
CORS	Flask-CORS	4.0	Allow browser API calls
Production Server	Gunicorn	Latest	WSGI server for Render
Frontend	Bootstrap 5	5.3.2	Responsive UI components
Frontend Icons	Bootstrap Icons	1.11.3	UI icon library
Version Control	Git + GitHub	—	Source code repository
Cloud Hosting	Render.com	—	Automatic deploy from GitHub
Model Storage	Google Drive	—	Host .pkl file for download